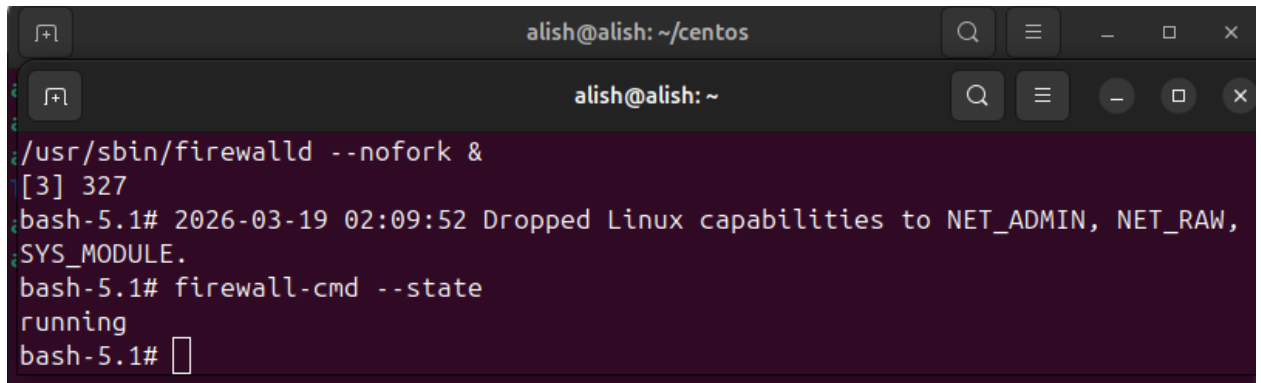


A. Install the firewall package and start the service

**Command :** – dnf install firewalld –y

– /usr/sbin/firewalld – –nofork &

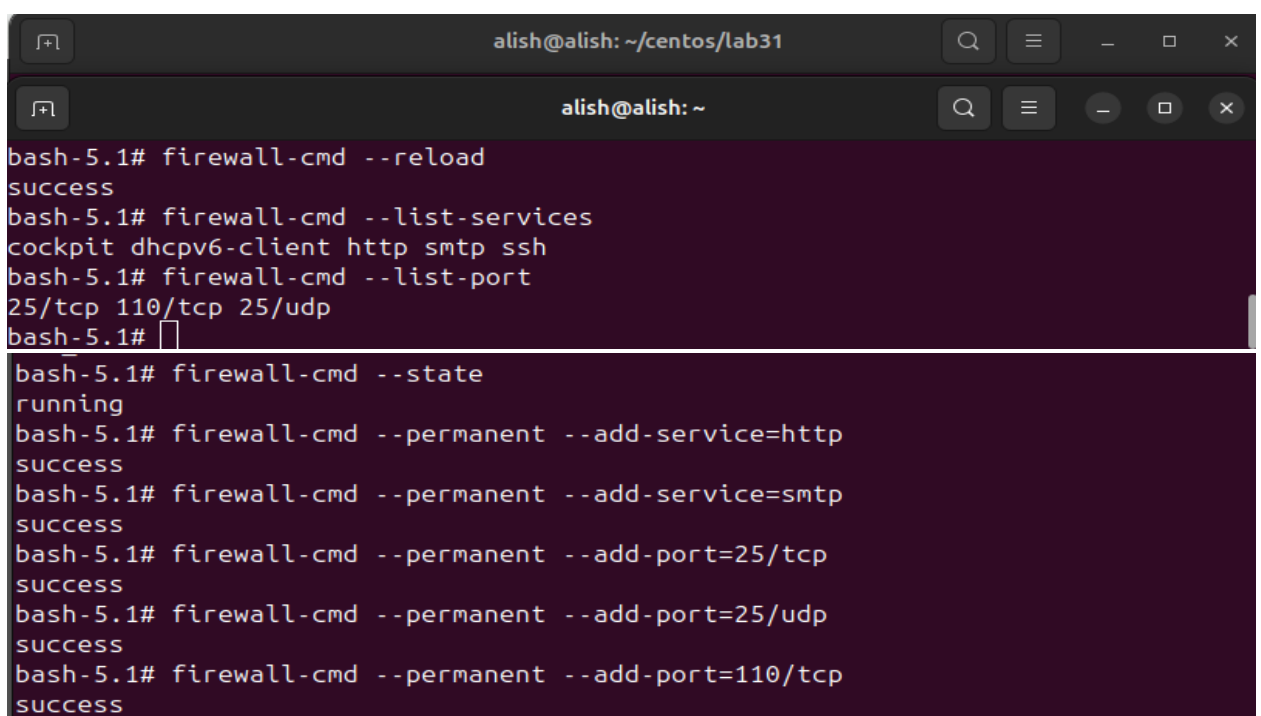
A terminal window with a dark background. The title bar shows 'alish@alish: ~/centos'. The prompt is 'alish@alish: ~'. The user enters the command '/usr/sbin/firewalld --nofork &'. The output shows '[3] 327' and then a message from bash-5.1: '2026-03-19 02:09:52 Dropped Linux capabilities to NET\_ADMIN, NET\_RAW, SYS\_MODULE.' followed by 'firewall-cmd --state' returning 'running'.

```
alish@alish: ~/centos
alish@alish: ~
/usr/sbin/firewalld --nofork &
[3] 327
bash-5.1# 2026-03-19 02:09:52 Dropped Linux capabilities to NET_ADMIN, NET_RAW,
SYS_MODULE.
bash-5.1# firewall-cmd --state
running
bash-5.1#
```

B. Add the following service and ports to allow packets through the firewall

[service =http,smtp] [port=25/tcp,25/udp,110/tcp]

**Command :** – firewall-cmd – –permanent – –add-service=http  
– firewall-cmd – –permanent – – add-service=smtp  
– firewall-cmd – – permanent – – add-port=25/tcp  
– firewall-cmd – – permanent – – add-port=25/udp  
– firewall-cmd – – permanent – – add-port=110/tcp  
– firewall-cmd – -reload  
– firewall-cmd – –list-services  
– firewall-cmd – –list-port

A terminal window with a dark background. The title bar shows 'alish@alish: ~/centos/lab31'. The prompt is 'alish@alish: ~'. The user enters several 'firewall-cmd' commands. The outputs show successful reload, listing of services (cockpit, dhcpv6-client, http, smtp, ssh) and ports (25/tcp, 110/tcp, 25/udp), and successful addition of http, smtp, and the three ports to the permanent firewall configuration.

```
alish@alish: ~/centos/lab31
alish@alish: ~
bash-5.1# firewall-cmd --reload
success
bash-5.1# firewall-cmd --list-services
cockpit dhcpv6-client http smtp ssh
bash-5.1# firewall-cmd --list-port
25/tcp 110/tcp 25/udp
bash-5.1#
bash-5.1# firewall-cmd --state
running
bash-5.1# firewall-cmd --permanent --add-service=http
success
bash-5.1# firewall-cmd --permanent --add-service=smtp
success
bash-5.1# firewall-cmd --permanent --add-port=25/tcp
success
bash-5.1# firewall-cmd --permanent --add-port=25/udp
success
bash-5.1# firewall-cmd --permanent --add-port=110/tcp
success
```

C. Remove the following service and port to block packets through the firewall

[service=smtp] [port=25/tcp,25/udp]

**Command:** – firewall-cmd - --permanent - -- remove-service=smtp

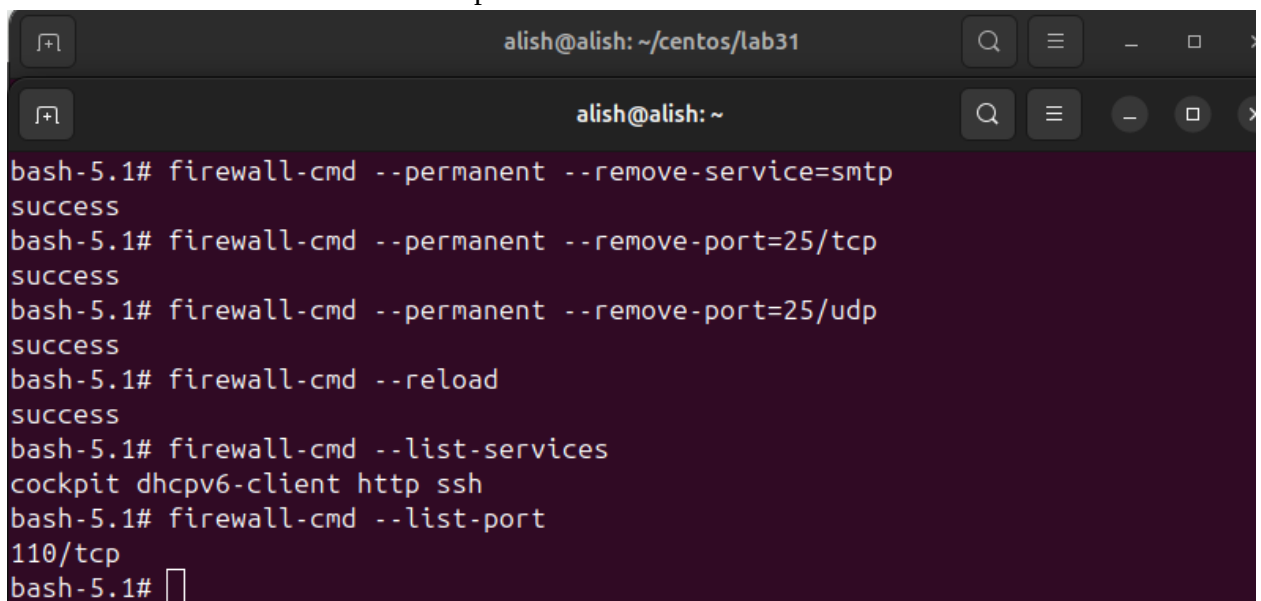
– firewall-cmd - --permanent - -- remove- port=25/tcp

– firewall-cmd - --permanent - --remove-port=25/udp

–firewall-cmd - -reload

– firewall-cmd - -- list-services

– firewall-cmd - --list-port

A screenshot of a terminal window with a dark background. The window title is 'alish@alish: ~/centos/lab31'. The terminal shows a series of commands and their outputs. The commands are: 'firewall-cmd --permanent --remove-service=smtp', 'firewall-cmd --permanent --remove-port=25/tcp', 'firewall-cmd --permanent --remove-port=25/udp', 'firewall-cmd --reload', 'firewall-cmd --list-services', and 'firewall-cmd --list-port'. The outputs are 'success', 'success', 'success', 'success', 'cockpit dhcpv6-client http ssh', and '110/tcp' respectively. The prompt 'bash-5.1#' is visible at the start of each command line.

```
alish@alish: ~/centos/lab31
bash-5.1# firewall-cmd --permanent --remove-service=smtp
success
bash-5.1# firewall-cmd --permanent --remove-port=25/tcp
success
bash-5.1# firewall-cmd --permanent --remove-port=25/udp
success
bash-5.1# firewall-cmd --reload
success
bash-5.1# firewall-cmd --list-services
cockpit dhcpv6-client http ssh
bash-5.1# firewall-cmd --list-port
110/tcp
bash-5.1#
```